

CRP-G4-CD60

Electric Cabinet Manual

This manual is applicable to RH series robot, such as RH14-10, RH18-20, RH20-10, etc. And for other robots, please refer to other relevant instructions.



Please ensure that the relevant instructions reach the end user of this product.

CROBOTP Relevant Instructions:

CRP Robot Safety Manual

Programming Instruction Manual

Robot Operation Manual

CRP-S80 PLC Manual

CRP-S100 PLC Manual

CRP-S80 System Hardware Manual

CRP-S100 System Hardware Manual

- Thank you very much for choosing our products !
- Please keep the relevant manuals of this product properly for reference when necessary !
- If the equipment needs to be transferred, please hand over the relevant information to the other party !
- Keys, functions and options not described in the robot related manual are deemed to be unavailable and should not be used !

Revision description:	
2022-09-05	First draft (From: NT-00031-A0)

Preface

1. Before using the robot, please be sure to carefully read the relevant instructions of our robot and operate the robot after understanding the content.

2 we solemnly suggest that all personnel involved in robot operation, teaching, maintenance, repair and spot inspection should study the operation instructions of our system in advance.

3. Our company reserves the right to change, revise or update this manual without prior notice.

4. This manual may not be reproduced or copied in whole or in part without the prior written permission of the company.

5. Please store this manual carefully to ensure that it reaches the end user. If the robot needs to be reinstalled, transported to different places, or sold to other users, please be sure to attach this manual. In case of loss or serious damage, please contact our agents or technicians.

6. All parameters, indicators and designs may be modified at any time without further notice without affecting the use effect.

7. We try to describe as many situations as possible in this manual. However, for those unnecessary and impossible situations, we don't describe them because of various possibilities. Therefore, those cases that aren't specifically described in the specification can be regarded as "impossible".

8. Omissions and errors are inevitable in the process of writing this book. If you find errors or incomprehensible places in the reading process, you are welcome to call for advice and correction.

Safety Instructions

Brief introduction

This section mainly introduces the safety principles and processes that need to be paid attention to when using robots. Before using robots, please be sure to read and understand the contents in this chapter, and operate robots according to the safety operation procedures. Before use (installation, operation, maintenance, overhaul), please be sure to read and master this manual and other relevant materials.

The charts, sequences and detailed explanations given in this manual may not be absolutely correct. Therefore, when using this manual to work, it is necessary to pay maximum attention. In case of any unspecified problems or troubles, please contact CRP.

In order to ensure the safety of each work, please read and fully understand the safety explanations and descriptions in this manual, the Robot Safety Manual, relevant laws, regulations, decrees and relevant materials, and take appropriate safety measures for each work.

In addition to the safety chapter, please note that there are other safety tips in the necessary part of the document.

Safety responsibility description

This manual doesn't apply to the robots which don't belong to our company. At the same time, our company will not be liable for accidents, damages and / or any problems related to industrial property rights that may be caused by the use of such robots.

We try our best to provide reliable safety information, but we will not be responsible for accidents or indirect accidents caused by the use of this manual and the products described therein.

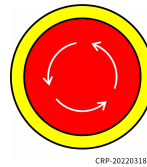
Except as expressly stated in this manual, the contents of this manual shall not be interpreted as any guarantee made by CRP for personal loss, property damage or specific applicability.

CRP isn't responsible for any possible errors in this manual.

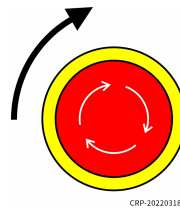
Emergency stop button

Emergency stop is a kind of safe stop, which is the highest priority function in robot system. Emergency stop buttons are installed on the teaching pendant, electric cabinet, station box, etc. In case of emergency, the user can press the emergency stop button to cut off the power supply of the robot immediately.

Most of the emergency stop buttons used for emergency stop use red main body, and the most common shape is mushroom head. As shown in the following figure.



If you need to reset, you need to rotate in the direction of the arrow on the emergency stop button (as shown in the figure below), and the emergency stop button will pop up and reset.



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1.Introduction to RH robot



CRP-RH series robots are hollow series robots developed by Chengdu CROBOTP Robot Technology Co., Ltd. It can be used in welding, handling, cutting, spraying, grinding and other related fields. CRP-G4-CD60 specification drive control integrated electrical control cabinet is adopted.

The main features of this series of robots are as follows:

- it is equipped with a safety emergency stop board independent of the control system. Imported forced disconnect relay is used to provide double circuit emergency stop externally to ensure the reliability of emergency stop.
- The function of automatic external power on : it can ensure the personal safety of the operator when the operator's line of sight shifts away from the robot.
- Energy saving mode: effectively reduce standby energy consumption. At the same time, avoid injury caused by personnel entering the robot area by mistake in the standby state.
- built in three-phase transformer, 380V and 200V isolation, more stable power supply. Built in three-phase filter can effectively isolate external interference and prevent internal interference output.
- Dual switching power supply design to avoid internal power interference.
- Special cables for high flexible robots are used for the body cables. Avoid failure caused by internal circuit damage.
- The body is provided with double circuit gas pipes to meet the connection requirements of welding and handling.

- 6-axis hollow large inner diameter, which can meet the installation requirements of water-cooled gun and bellows gun. At the same time, avoid the entanglement of handling fixture pipeline and cable.
- Welding cable can be built in to facilitate the use of welding robot.

2. Description of G4 control cabinet (electric cabinet)

2.1 Introduction to control cabinet



CRP-G4-CD60 control cabinet has the following characteristics:

- Adopt split design; It is divided into power supply room and control room. The power supply room mainly focuses on heating devices, and adopts a multi group fan + air duct design to ensure good heat dissipation of heating devices. The control room is isolated from the power supply room to prevent dust and oil from entering the control room, ensure the cleanliness of the control room, and avoid abnormal operation of components in the control room affected by dust and oil.
- Multiple sets of anti-interference components: power filter, isolation transformer, dual switching power supply. Ensure the stable and reliable operation of the control cabinet.
- The built-in safety emergency stop board adopts double safety relays.

When any emergency stop is pressed, the strong current circuit and the servo unit deal with the emergency at the same time and forcibly stop the robot. Thereby achieving a safe emergency stop.

- collision softening: when the anti-collision signal is effective, the system and the drive respond at the same time to quickly soften the attitude axis and avoid damaging the body and welding gun.

2.2 Specification parameters of control cabinet

Teach Pendant	8-inch TFT-LCD, keyboard + touch screen, mode selection switch, safety switch, emergency stop button
User memory	200M
Axis number	6+2 aixs , the standard configuration is 6 axes, and the external 2 axes needs to be selected)
Interface	Digital I/O interface,24 input/output(expandable COM)
	4 channels of 0-10V analog output, 12 bit accuracy, com expandable
	Dual encoder signal interface (for position tracking)
	Ethernet interface
	Dual USB interface
Operation mode	Teach, Play, Remote
Path of movement	Point to point, Line, Arc, Circle
Instruction system	Move, Logic, Process, Arithmetic
Coordinate system	Joint coordinate, Rrectangular coordinate, User coordinate, Tool coordinate, Base coordinate
Abnormal etection function	Emergency stop abnormal,servo abnormal,user coordinate abnormal, tool coordinates abnormal,safety maintenance,arcing abnormal, etc
Robot safety	External emergency stop, anti-collision, safety latch and other interfaces; MC safety circuit, servo softening, etc.
Reserved special interface:	Special interface for arc welding, station interface and remote upper interface
Software package	welding,handling,palletizing, painting ,bending, stamping , etc (optional)

Other	Built in PLC, power-off regeneration, encoder interface (supporting synchronous belt), arc tracking and parts (optional), visual software (optional), laser tracking software (optional), etc.
Connection cable	3m
Power supply	AC 3-phase 380V 50HZ
Dimensions (MM)	550 x 785 x 410
Weight (KG)	65KG (Without extension)

2.3 Installation dimension of electric cabinet

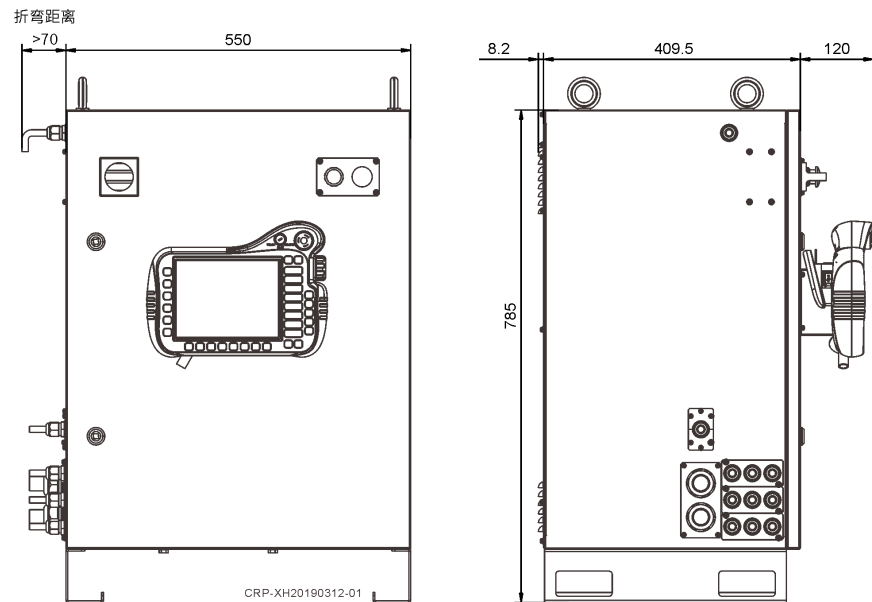


Fig.2.3.1

2.4 Installation requirements of electric cabinet

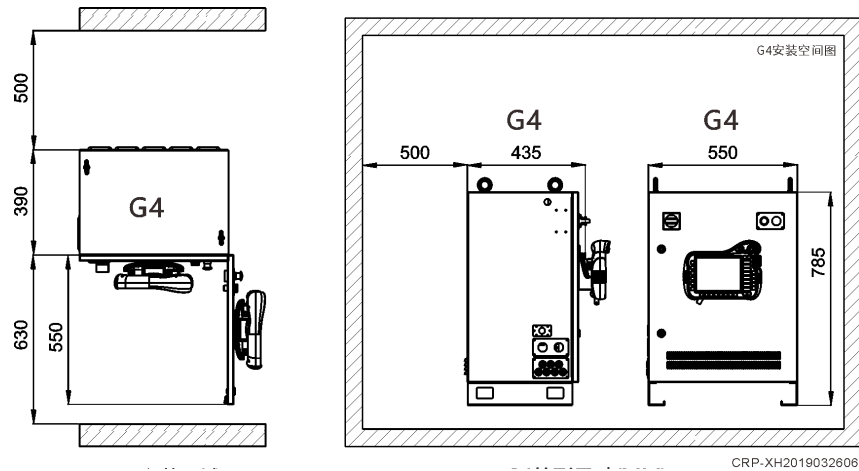


Fig.2.4.1

Please install CRP-G4-CD60 control cabinet according to the size shown above. Ensure that there is enough space for the maintenance of robots, G4 cabinets and peripheral equipment. Otherwise, unexpected injuries may occur during maintenance.



CAUTION

Be careful when lifting the robot. Due to the different posture of the robot and the installation status of the option class, the robot will tilt forward / backward. If the robot is lifted in an inclined state, any impact will shake the robot and damage the surrounding objects. After handling, please remove the lifting ring installed on the robot.

The correct posture of robot handling is shown in the figure below.

3. Electrical Cabinet Layout and Interface Description

3.1 Electrical cabinet layout

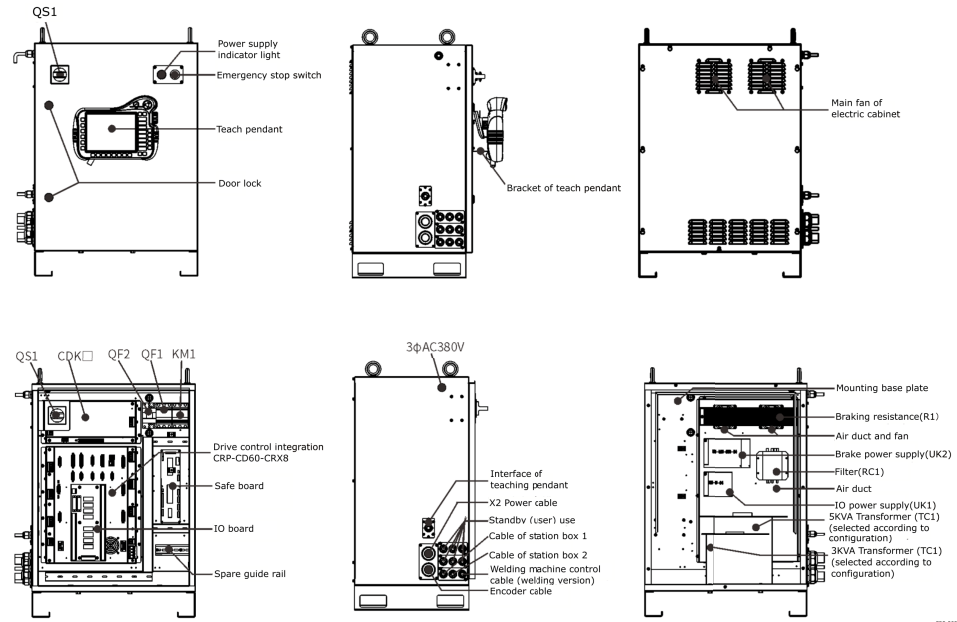


Fig.3.1.1

3.2 Main components

3.2.1 Power supply

The power supply of this electric cabinet is powered by 3 ϕ AC380V, $\pm 10\%$, 50Hz power supply. The power capacity is selected according to the configured transformer capacity.

When using transformer with extended configuration (5KVA). External power supply capacity shall be $> 5\text{KVA}$, Wire diameter $\geq 4\text{mm}^2$.

When using transformer with extended configuration (8KVA). External power supply capacity shall be $> 8\text{KVA}$, Wire diameter $\geq 4\text{mm}^2$.

When an external transformer is used, the power supply capacity and wire diameter shall be selected according to the capacity of the external transformer.



WARNING

The power supply of the electric cabinet must be reliably grounded, and the wrong side will cause equipment failure or accident.

3.2.2 Main switch (QS1)



Fig.3.2.1

When the switch is on, the strong current in the control cabinet is turned on and the power indicator is on.

Specification and model: KCF1PZC + V0C (25A)



WARNING

Non professionals are forbidden to open the control cabinet door. When it is really necessary to open the control cabinet door, be sure to cut off the power switch or operate carefully before opening the control cabinet door. Otherwise, personal injury or equipment damage may be caused.

3.2.3 Fuse (QF2)



Fig.3.2.2

The main function of this fuse is to protect the power supply of the control circuit of the drive control integrated unit, the main fan of the electric cabinet, the brake power supply and the IO power supply. When the indicator light is on, it means that the fuse is blown. At this time, it is necessary to check the relevant circuit first, and then reinstall the fuse to close after confirming that the circuit is normal.

Specification and model: RT28N-32X / 3P + RT18-32 (6A)

3.2.4 Air switch (QF1)



Fig.3.2.3

This device is mainly used for the power supply circuit of the protection drive control integrated unit (CD60-CRX8).

Specification and model: IC65N-3P-D25A

3.2.5 Contactor (KM1)



Fig.3.2.4

The main function of this device is to connect or disconnect the power electric circuit of the drive control integrated unit (CD60-CRX8) in cooperation with the safety emergency stop plate.

Specification and model: LC1D25M7C

3.2.6 Integrated drive control unit (CRP-CD60-CRX8)



Fig.3.2.5

This drive control integrated unit (CD60-CRX8) is the core component of this control cabinet. This unit includes a robot control unit and a robot drive unit. It is used for all functions and motion control of the robot, so please take care of it.

Please refer to [CRP-CD60-CRX8 Operation Manual](#) for detailed operation instructions and interface instructions of this drive control integrated unit.

3.2.7 Expansion axis (CDK2)

This expansion axis module is used when the robot needs to expand the external 7th axis, or the 7th and 8th axes. Please refer to [CRP-CD60-CRX8 System Hardware Manual](#) for operating instructions and interfaces.

3.2.8 Safe Board



Fig.3.2.6

The safety emergency stop board adopts double emergency stop circuit access, with two built-in safety relays with safety certification, and all emergency stop signals are processed by the associated box. When any emergency stop is pressed, the strong current circuit and the servo unit will deal with the emergency at the same time and forcibly stop the robot. It has a high safety level and can effectively protect personnel and equipment from injury.

3.2.9 I / O adapter board (for users)

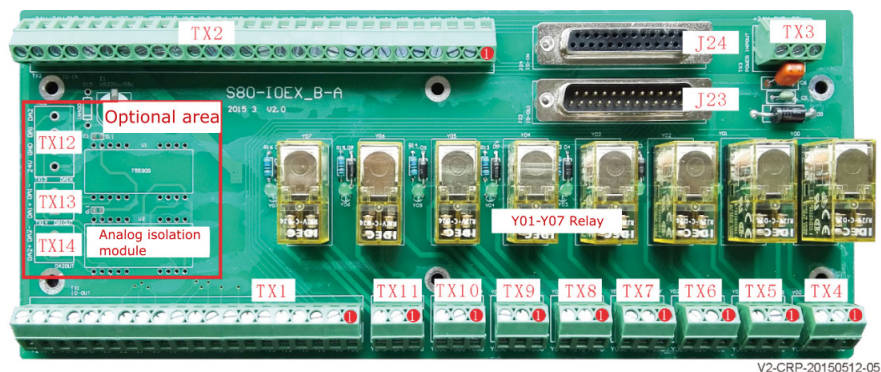


Fig.3.2.7

The I / O adapter board is mainly used to connect external IO signals to x00-x22 and y00-y22 to interact with the control unit. Among them, y00-y07 carries out relay transfer output, and the maximum capacity of relay contact is 2A. The optional area is the analog isolation part. When the analog quantity is used to connect the welding machine, this part of the circuit can be used after isolating two chanal analog quantities.

-
1. The TX terminal of the board must be connected with H-shaped terminal and pressed with special pliers. Prevent loose wiring.

2. The + 24V and GND signals on TX1 and TX2 terminals are connected with the + 24V and GND signals of tx3, which are used for external switches, electrical appliances and other equipment.

1. J24 IO-IN pin definition

Function: J24 is connected to the input interface of the host through the matching 'I / O signal input cable'			
Pin	Name	Definition	Valid status
1	X00	Input port	Low level (0V) valid
2	X02	Input port	Low level (0V) valid

3	X04	Input port	Low level (0V) valid
4	X06	Input port	Low level (0V) valid
5	X08	Input port	Low level (0V) valid
6	X10	Input port	Low level (0V) valid
7	X12	Input port	Low level (0V) valid
8	X14	Input port	Low level (0V) valid
9	X16	Input port	Low level (0V) valid
10	X18	Input port	Low level (0V) valid
11	X20	Input port	Low level (0V) valid
12	X22	Input port	Low level (0V) valid
13	GND	Ground wire: 0V	Supply power to the system
14	X01	Input port	Low level (0V) valid
15	X03	Input port	Low level (0V) valid
16	X05	Input port	Low level (0V) valid
17	X07	Input port	Low level (0V) valid
18	X09	Input port	Low level (0V) valid
19	X11	Input port	Low level (0V) valid
20	X13	Input port	Low level (0V) valid
21	X15	Input port	Low level (0V) valid
22	X17	Input port	Low level (0V) valid
23	X19	Input port	Low level (0V) valid
24	X21	Input port	Low level (0V) valid
25	GND	Ground wire: 0V	Supply power to the system

2. J24 IO-OUT pin definition

Function: J23 is connected to the input interface of the host through the matching 'I / O signal output cable'			
Pin	Name	Definition	Valid status
1	Y00	Output port	Low level (0V) valid
2	Y02	Output port	Low level (0V) valid
3	Y04	Output port	Low level (0V) valid
4	Y06	Output port	Low level (0V) valid
5	Y08	Output port	Low level (0V) valid
6	Y10	Output port	Low level (0V) valid
7	Y12	Output port	Low level (0V) valid
8	Y14	Output port	Low level (0V) valid
9	Y16	Output port	Low level (0V) valid
10	Y18	Output port	Low level (0V) valid
11	Y20	Output port	Low level (0V) valid
12	Y22	Output port	Low level (0V) valid

13	+24V	24V power supply	Supply power to the system
14	Y01	Output port	Low level (0V) valid
15	Y03	Output port	Low level (0V) valid
16	Y05	Output port	Low level (0V) valid
17	Y07	Output port	Low level (0V) valid
18	Y09	Output port	Low level (0V) valid
19	Y11	Output port	Low level (0V) valid
20	Y13	Output port	Low level (0V) valid
21	Y15	Output port	Low level (0V) valid
22	Y17	Output port	Low level (0V) valid
23	Y19	Output port	Low level (0V) valid
24	Y21	Output port	Low level (0V) valid
25	+24V	24V power supply	Supply power to the system

3. TX1 IO-OUT terminal definition

TX1 output signal			
Pin port	Name	Definition	Valid status
1	Y08	Output port	Low level (0V) valid, maximum output current 200mA
2	Y09	Output port	Low level (0V) valid, maximum output current 200mA
3	Y10	Output port	Low level (0V) valid, maximum output current 200mA
4	Y11	Output port	Low level (0V) valid, maximum output current 200mA
5	Y12	Output port	Low level (0V) valid, maximum output current 200mA
6	Y13	Output port	Low level (0V) valid, maximum output current 200mA
7	Y14	Output port	Low level (0V) valid, maximum output current 200mA
8	Y15	Output port	Low level (0V) valid, maximum output current 200mA
9	Y16	Output port	Low level (0V) valid, maximum output current 200mA
10	Y17	Output port	Low level (0V) valid, maximum output current 200mA
11	Y18	Output port	Low level (0V) valid, maximum output current 200mA
12	Y19	Output port	Low level (0V) valid, maximum output current 200mA

13	Y20	Output port	Low level (0V) valid, maximum output current 200mA
14	Y21	Output port	Low level (0V) valid, maximum output current 200mA
15	Y22	Output port	Low level (0V) valid, maximum output current 200mA
16			
17	GND	Ground wire: 0V	
18	GND	Ground wire: 0V	
19	+24V	Supply power: +24V	
20	+24V	Out +24V	

Schematic diagram of output interface Y00-Y22:

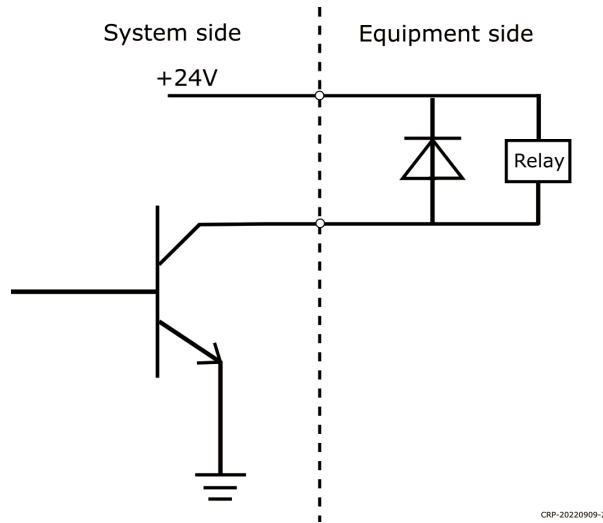


Fig.3.2.8

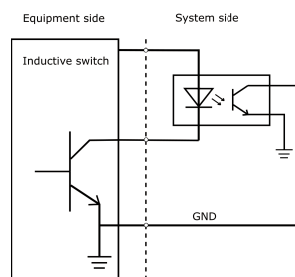
★Note

1. The internal of the system is the open collector output of the transistor.
2. The maximum load current is 100mA.
3. Free wheeling diodes must be connected when controlling inductive loads (such as relays).
4. The station box 2 occupies y21 and Y22 interfaces. When these two interfaces need to be used for other purposes, the corresponding PLC circuit can be adjusted. Shield the station box 2 and release the two interfaces.

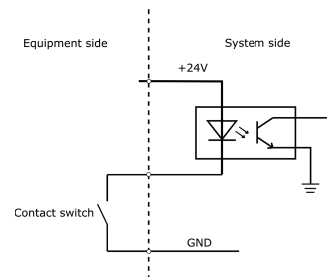
4. TX2 IO-IN Pin definition

TX2 Input terminal			
Terminal pin	Name	Definition	Valid status
1	X00	Input port	Low level (0V) valid
2	X01	Input port	Low level (0V) valid
3	X02	Input port	Low level (0V) valid
4	X03	Input port	Low level (0V) valid
5	X04	Input port	Low level (0V) valid
6	X05	Input port	Low level (0V) valid
7	X06	Input port	Low level (0V) valid
8	X07	Input port	Low level (0V) valid
9	X08	Input port	Low level (0V) valid
10	X09	Input port	Low level (0V) valid
11	X10	Input port	Low level (0V) valid
12	X11	Input port	Low level (0V) valid
13	X12	Input port	Low level (0V) valid
14	X13	Input port	Low level (0V) valid
15	X14	Input port	Low level (0V) valid
16	X15	Input port	Low level (0V) valid
17	X16	Input port	Low level (0V) valid
18	X17	Input port	Low level (0V) valid
19	X18	Input port	Low level (0V) valid
20	X19	Input port	Low level (0V) valid
21	X20	Input port	Low level (0V) valid
22	X21	Input port	Low level (0V) valid
23	X22	Input port	Low level (0V) valid
24			
25	GND	Ground wire: 0V	
26	GND	Ground wire: 0V	
27	+24V	Out +24V	
28	+24V	Out +24V	

Schematic diagram of input interface principle of X00-X22:



Example of inductive switch



Example of contact switch

Fig.3.2.9

★Note

1. Whether the contact of the switch is normally open or normally closed depends on the interface definition.
2. The capacity of the switch shall not be less than 16mA.
3. NPN type shall be selected when selecting inductive switch.
4. Station box 2 occupies X21 and X22 interfaces. When these two interfaces need to be used for other purposes, the corresponding PLC circuit can be adjusted. Shield station box 2 and release these two interfaces. Refer to "5.3.3 release of station box 2".

5. Definition of TX3 power input power terminal

TX3 I / O interface power supply			
Terminal pin	Name	Definition	Valid status
1	+24V	Positive port of I/O interface power supply	\
2	GND	Negative port of I / O interface power supply	\
3	PE	Grounding	\

★Note

The I / O interface power supply is the 24V input power supply of the IO board, and the external power supply module is used for power supply.

6. TX4 Y00 terminal definition

TX4 Y00			
Terminal pin	Name	Definition	Valid status
1	Y00O	Y00 relay output is normally open	\
2	Y00G	Common terminal of Y00 relay output	\
3	Y00C	Y00 relay output is normally closed	\

7. Definition of TX5 Y01 terminal

TX5 Y01			
Terminal pin	Name	Definition	Valid status
1	Y01O	Y01 relay output is normally open	\
2	Y01G	Common terminal of Y01 relay output	\
3	Y01C	Y01 relay output is normally closed	\

8. TX6 Y02 terminal definition

TX6 Y02			
Terminal pin	Name	Definition	Valid status
1	Y02O	Y02 relay output is normally open	\
2	Y02G	Common terminal of Y02 relay output	\
3	Y02C	Y02 relay output is normally closed	\

9. TX7 Y032 terminal definition

TX7 Y03			
Terminal pin	Name	Definition	Valid status
1	Y03O	Y03 relay output is normally open	\
2	Y03G	Common terminal of Y03 relay output	\
3	Y03C	Y03 relay output is normally closed	\

10. TX8 Y04 terminal definition

TX8 Y04			
Terminal pin	Name	Definition	Valid status
1	Y04O	Y04 relay output is normally open	\
2	Y04G	Common terminal of Y04 relay output	\
3	Y04C	Y04 relay output is normally closed	\

11. TX9 Y05 terminal definition

TX9 Y05			
Terminal pin	Name	Definition	Valid status
1	Y05O	Y05 relay output is normally open	\
2	Y05G	Common terminal of Y05 relay output	\
3	Y05C	Y05 relay output is normally closed	\

12. TX10 Y06 terminal definition

TX10 Y06			
Terminal pin	Name	Definition	Valid status
1	Y06O	Y06 relay output is normally open	\
2	Y06G	Common terminal of Y06 relay output	\
3	Y06C	Y06 relay output is normally closed	\

13. TX11 Y07 terminal definition

TX11 Y07			
Terminal pin	Name	Definition	Valid status
1	Y07O	Y07 relay output is normally open	\
2	Y07G	Common terminal of Y07 relay output	\
3	Y07C	Y07 relay output is normally closed	\

14. Analog isolation board

★Note
1. Analog isolation plate is the standard configuration of welding robot. The carrying robot is empty. 2. The analog isolation board is an optional part, which is used to isolate the system analog quantity and the input analog quantity of the welding machine to prevent interference! 3. When the analog isolation board is not selected, the customer can also directly connect the system AVO to the input terminal of the welding machine for testing. In formal use, it is strongly recommended to use analog isolation board.

15. TX12 terminal definition

TX12 Analog quantity input terminal, connected with AVO interface of host			
Terminal pin	Name	Definition	Valid status
1	24V	+24V Input	\
2	GND	0V Input	\
3	DA1	DA1 Analog input	\
4	DA2	DA2 Analog input	

16. TX13 terminal definition

TX13 Connecting welding current control terminal			
Terminal pin	Name	Definition	Valid status
1	DA1+	Output positive terminal of welding current	\
2	DA1-	Output negative terminal of welding current	\

★Note

Connect the current control terminal of the welding machine to provide current control for the welding machine. It is the output port of the isolation plate.

17. TX14 terminal definition

TX14 Connect the welding voltage control terminal			
Terminal pin	Name	Definition	Valid status
1	DA2+	Output positive terminal of welding voltage	\
2	DA2-	Output negative terminal of welding voltage	\

★Note

Connect the voltage control terminal of the welding machine to provide voltage control for the welding machine. It is the output port of the isolation plate.

3.2.10 Spare guide rail

This guide rail is used for the reserved installation position when the customer needs to add relays or other components.

3.2.11 Braking resistance (R1)

Braking resistance is also called regenerative resistance. It is used when the servo motor starts to consume excess energy when the generated energy exceeds the energy charged by the capacitor in the generator mode.

The generator mode includes the following conditions:

1. During deceleration stop during acceleration and deceleration operation.
2. The servo motor formed by the load side operates continuously without interruption (load).

Specification and model: 800W 20RJ

★Note
1. Since the braking resistor is mainly used for energy consumption, it will generate a lot of heat according to the movement of the robot, so it is strictly prohibited to touch it when it is powered on. 2. Good heat dissipation conditions shall be guaranteed due to heat generation. 3. After the main switch is turned off, wait for more than 5 minutes before touching the surface of the brake resistor.

3.2.12 Air duct / fan

During the operation of the integrated drive control unit, its power devices will generate a large amount of heat, which is mainly conducted to the radiator of the integrated drive control unit. If the heat cannot be effectively dissipated, the integrated drive control unit will give an over temperature alarm. And the air duct and the air duct fan are specially used to dissipate this part of heat.

At the same time, the air duct fan is also used to dissipate the heat of the braking resistor.

Specification and model: ME80252V1-000C-A99

★Note
Air ducts and fans need regular inspection and maintenance to prevent blockage, failure and influence on heat dissipation effect.

3.2.13 UK1/UK2/UK3

The band brake power supply UK2 / UK3 is mainly used to supply power to the robot body and external axis servo motors. IO power supply UK1 is mainly used to supply power to I / O board. Double switching power supply design is used to avoid interference to IO signal and control system when holding brake operates.

Specification and model: UK1: LRS-50-24 UK2: LRS-100-24 UK3: LRS-100-24

3.2.14 Filter (RC1)

It is mainly used to power the drive control integrated unit.

The selection of transformer capacity is directly related to the load. The control cabinet adopts a three-phase isolation transformer to avoid the danger of electric shock caused by any line at the output end contacting with people and ensure the safety of electricity; At the same time, the input end and the output end are completely isolated, which can filter the input power supply well and prevent interference.

The control cabinet is currently configured with the following specifications:

SG5000VA: for RH14-10 with external shaft.

SG8000VA: for RH14-10 with external shaft.

★Note
1. When the transformer is working, the input terminal still carries strong current, so it is forbidden to touch it when it is energized to prevent electric shock. 2. Heat will be generated during the operation of the transformer, so it is strictly prohibited to touch it when it is powered on to prevent scalding. At the same time, good ventilation conditions shall be ensured.

3.2.15 Main fan of electric cabinet

There are transformers, radiators of integrated drive control units, braking resistors and other heating devices in the power supply room; If the robot can not get good heat dissipation, it will lead to a series of abnormalities such as abnormal operation of the robot and alarm. The main fan of the electrical cabinet is used for ventilation and heat dissipation of the power supply room. In order to ensure a good heat dissipation effect, it is necessary to regularly check the operation of the fan and the blockage of the air outlet.

Specification and model: DP200A 2123XBL

3.2.16 Teaching pendant

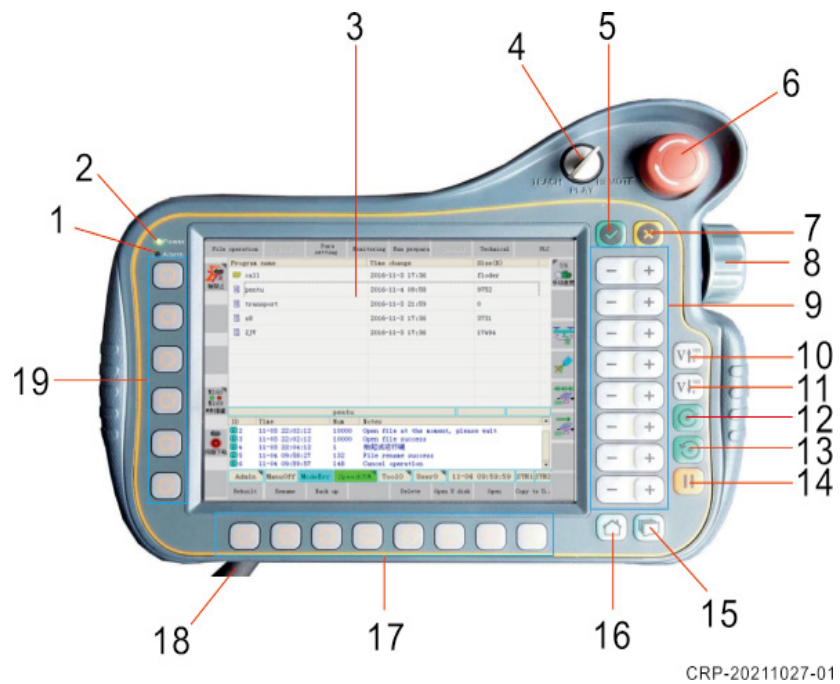


Fig.3.2.10 Front of touch screen teaching pendant (V2)

- | | |
|---|--------------------------------------|
| 1. Alarm indicator | 2. Power indicator |
| 3. Display / touch screen | 4. Mode selection (key switch) |
| 5. Confirm key | 6. Emergency stop button |
| 7. Cancel key | 8. Handwheel |
| 9. Movement keys of each axis | 10. Speed magnification increase Key |
| 11. Speed magnification reduction key | |
| 12. Test run and forward run key | |
| 13. Reverse operation is standby, and the function is the same as forward running | |
| 14. Pause key | 15. Window switch key |
| 16. Home key (standby) | 17. Submenu key |
| 18. Teaching pendant cable | 19. Status control Key |



Fig.3.2.11 Back of touch screen teaching pendant (V2)

- | | |
|------------------|-------------|
| 1. Touch pen | 2. Handrail |
| 3. Safety switch | 4. USB |

The teaching device is a robot control operator, mainly used for teaching programming, state monitoring, operation preparation and other operations. Built in LCD display and touch screen, these components are fragile components, please handle them with care and use them with care.

Please refer to CRP (touch screen version) Operation Manual for detailed introduction and operation method of the teaching pendant.

3.2.17 Station box

The station box is mainly used for play, remote or reservation servo power on and program start and stop.

For detailed description, please refer to "5.3 detailed description of station box".

4. Instructions for use of electric cabinet

4.1 Power on steps of electric cabinet

Power on the control cabinet according to the following steps:

1. The user's power distribution is powered on.
2. Turn the main switch (QS1) to the on position.
3. Wait for about 70 seconds, and the system starts up and displays the program list interface.

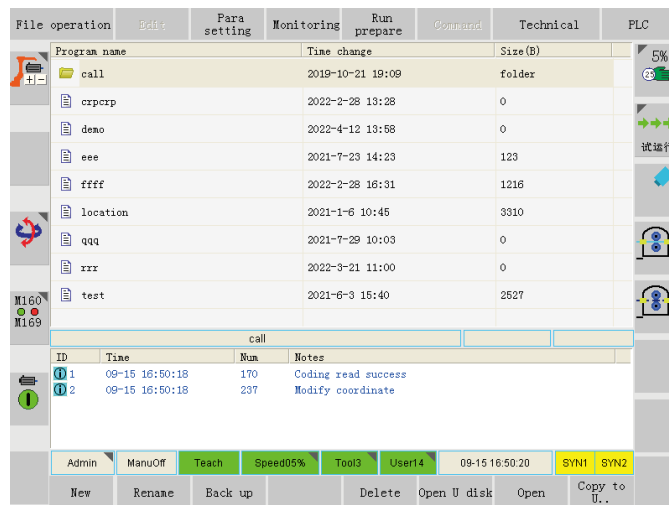


Fig.4.1.1

4. Press the < servo power down > button, and the system reads the absolute position. Display < servo power on > after completion.
5. At this time, the system is powered on and can be used for subsequent work.

4.2 Servo power on in play or remote mode

When the key switch is switched to play or remote mode, the system will automatically switch the servo power on state to servo power off. The servo cannot be powered on through the < servo power off > key or the corresponding physical key on the screen. At this time, the servo must only be powered on by pressing the servo power on (orange button) of the station box 1. When the servo is powered on, the orange indicator light of servo power on of station box 1 is on.

For details of the station box, please refer to "5.3 detailed introduction of the station box".

★Note

1. This function is forced by the system. Please don't try to turn it off.
2. The main purpose of this function is for secondary confirmation, so as to avoid personal injury caused by operators only focusing on the start-up procedure of the teaching box and ignoring the position and status of the robot. Meanwhile, relevant national standards and Safety specifications shall be followed.

J11 (emergency stop interface of station box 1), j12 (emergency stop interface of station box 2), j13 (emergency stop interface of station box 3), J14 (emergency stop interface of station box 4) and j13 (standby emergency stop interface) among them, there are four terminals in each group, of which 1 and 2 are a group and 3 and 4 are a group. Note that they cannot be cross connected and grounded, otherwise the emergency stop of this circuit will not work.

5.1.2 Troubleshooting of safety emergency stop plate

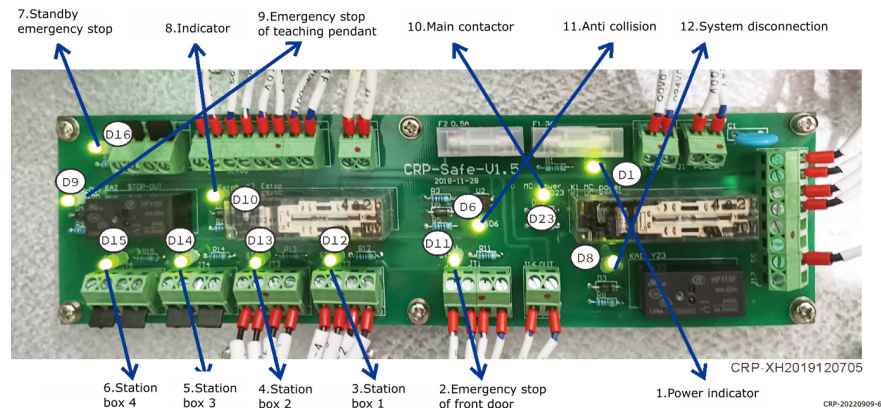


Fig.5.1.2

After all emergency stop buttons (E-stop of teaching box, E-stop of station box 1, E-stop of station box 2, E-stop of electric cabinet door, etc.) are pressed, the robot will stop quickly, the teaching pendant will report "hardware emergency stop". After resetting the emergency stop switch, press the reset key "R" or the servo power on button of station box 1 to reset the alarm. If "hardware emergency stop" is always reported, you can check the indicator light of the emergency stop board in the electrical cabinet to check whether the corresponding circuit is abnormal. According to the sequence of lights in the above figure, dispose of the lights that are not lit from ① (when the circuit is normal, the lights are on).

5.2 Energy saving mode

5.2.1 Introduction

Energy saving mode means that when the robot is in the enabled state, the system is not in the wait state, and the system does not send pulses to control the motor to run, if it exceeds 1 minute, the system will automatically disable it, and the motor will brake.

Main uses:

Safe :When the conditions for the energy-saving mode are met, the robot is quiet and still. and the operator enters the working area because they think that the robot has stopped; However, at this time, the robot is still enabled. Once the motion is triggered, the robot will start to operate immediately, which will cause injury to personnel or equipment. The timeout is automatically disabled, which can well avoid this risk.

Energy saving: when the servo is enabled, the motor brake need energy to drive. After the disable, the brake is locked (restored to the original position, the robot joint is locked), and the motor is in a free state, no longer consuming energy.

5.2.2 Energy saving mode adjustment

When the customer uses external equipment (PLC, etc.) to control the start and stop of the robot on site, there may be timeout, too short time, or conflict with the energy saving mode. Please contact our technical personnel at this time.



WARNING

Based on the safe use of the energy-saving mode, do not adjust the energy-saving mode unless it is absolutely necessary.

5.3 Station box details

5.3.1 Description of station box

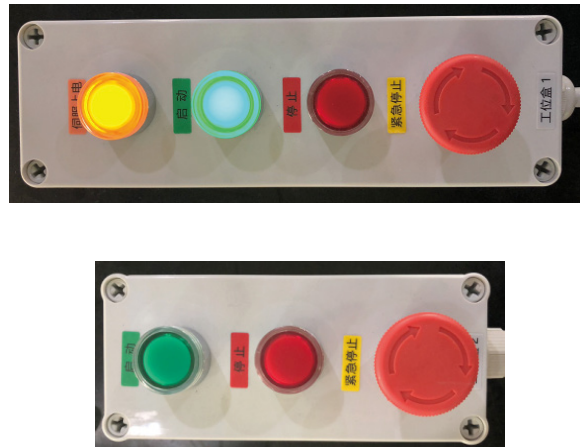


Fig.5.3.1

Station box 1 (station 1) has four operation switches, which are mainly used for servo power on of play or remote mode; Program start and stop of play mode; Program start and stop of remote mode; Station 1 in reservation mode starts and stops. Station box 2 (station 2; configured for welding version, but not for transport version) has three operation switches, which are mainly used to start and stop station 2 in reservation mode.

The functions of each operation switch are as follows:

- servo power on (orange button): used when the user powers on the servo in reproduction mode or remote mode. When the teaching pendant is switched to play or teaching mode, the robot system will automatically servo power down. At this time, this button must be pressed to servo power up. When the servo is powered on, the orange indicator of this button will be on.



WARNING

1. This function is forced by the system. Please do not try to turn it off.
2. The main purpose of this function is for secondary confirmation, so as to avoid personal injury caused by operators only focusing on the start-up procedure of the teaching pendant and ignoring the position and status of the robot. Meanwhile, relevant national standards and Safety specifications shall be abided.

- Start button (green button): it is used to start the program of the corresponding mode and station. When the program starts to run. The green indicator light of this button is on.

- Stop button (red button): used to stop the running program. When the program stops, the red indicator light of this button will be on.

- Emergency stop button (mushroom head button): used to stop the robot in case of abnormality.

5.3.2 Definition of station box and interface

1. Servo power on button and light of station box 1 (Station 1).

In the teaching mode, the lamp lights up after enabling, (Y61).

In reproduction and remote mode, press this button to clear the alarm and enable (X86).

2. Start button and light of station box 1.

In play mode, after enabling, press this button (X84) to start the current program, and the light is on (Y60).

In remote mode, after enabling, press this button to start the specified program, and the light is on (Y60). In case of reservation, the light flashes.

3. Stop button and light of station box 1.

In play and remote mode, press this button (x85) during operation to slow down and stop the current program, and the light is on (y62). If the robot is stopped, the light is on.

4. Start button and light of station box 2 (station 2)

In the remote mode, after enabling, press this button (X21) to start the specified program, and the light is on (y21). If it is a reservation, the light flashes.

5. Stop button and lamp of station box 2

In play and remote mode, press this button (X22) during operation to slow down and stop the current program, and the light is on (Y22). If the robot is stopped, the light is on.

5.3.3 Release of station box 2

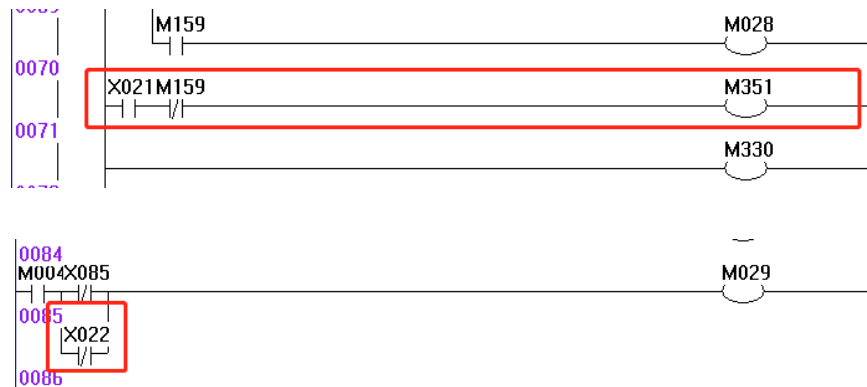
The content of this subsection is only for the station box 2, which does not need to be used; When there are two input and output ports at the same time, use this section to release the X21, X22, y21 and Y22 interfaces occupied by the station box 2. This subsection is not applicable to other occasions.



WARNING

Since this section involves system PLC, please handle it carefully.

1. Modify the system PLC and delete the following circuits in the red box.



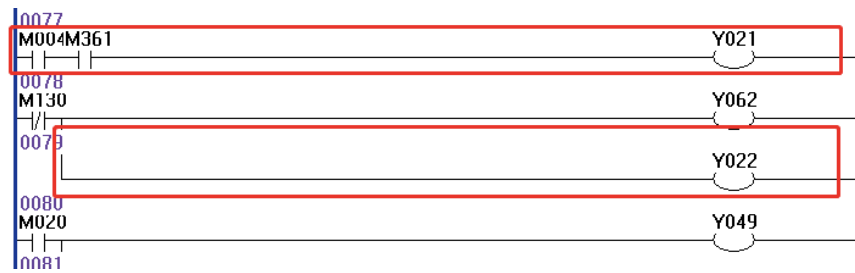


Fig.5.3.2

2. Disconnect the physical wiring of X21, X22, y21 and Y22. It is recommended not to remove the emergency stop signal line of station box 2. Station box 2 is used as an emergency stop box.

3. If the emergency stop of station box 2 still needs to be removed, the corresponding terminal shall be short circuited according to the "short circuiting position" as shown in the figure below.

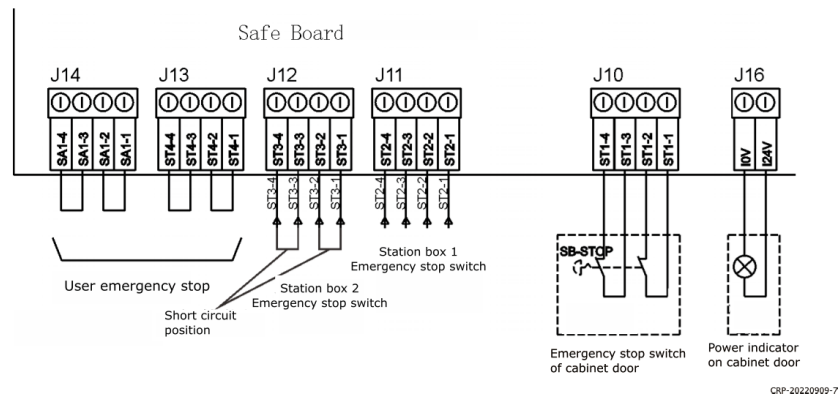


Fig.5.3.3

5.3.4 Add station box

The content of this section is for the occasions where the two station boxes are not enough and the station boxes need to be added. The system supports up to 10 station boxes.

Example: add three more station boxes.

Since the system comes with two station boxes, so add station box 3 (corresponding to station 3), station box 4 (corresponding to station 4) and station box 5 (corresponding to station 5) are added. and so on.

For detailed introduction of PLC, please refer to [CRX8 System PLC Manual](#).

Definition of auxiliary relay:

Definition of auxiliary relay:	definition	Type	illustration
M62	Positive hard limit is valid	Input	
M63	Negative hard limit is valid	Input	
M66	drive alarm of J9 axis	Input	1: drive alarm of J9-J12 axis; 1: drive ready of J9-J12axis ;
M67	drive ready of J9 axis	Input	
M68	drive alarm of J10 axis	Input	
M69	drive ready of J10 axis	Input	
M70	drive alarm of J11 axis	Input	
M71	drive ready of J11 axis	Input	
M72	drive alarm of J12 axis	Input	
M73	drive ready of J12 axis	Input	
M74	Servo enable of J9 axis is completed	Input	1: Servo enable of J9-J12 axis is completed
M75	JServo enable of J10 axis is completed	Input	
M76	Servo enable of J11 axis is completed	Input	
M77	Servo enable of J12 axis is completed	Input	
M298	J9 axis interference	status	1: J9-j10 axis enters the interference area;
M299	J10 axis interference	status	0: outside the interference area;

M330	Reservation running permission signal of station 1	Input	The reserved running permission signal of station 1 is detected before the reserved program starts to run. The rest of the time is not detected. 1 - allowed 0 - not allowed
M331	Reservation running permission signal of station 2	Input	
M332	Reservation running permission signal of station 3	Input	
M333	Reservation running permission signal of station 4	Input	
M334	Reservation running permission signal of station 5	Input	
M335	Reservation running permission signal of station 6	Input	
M336	Reservation running permission signal of station 7	Input	
M337	Reservation running permission signal of station 8	Input	
M338	Reservation running permission signal of station 9	Input	
M339	Reservation running permission signal of station 10	Input	
M340	Reservation 0 of station fault	Input	The signal must be normal all the time, otherwise the reservation cannot be made when the reservation is required, the alarm stops during the reservation, and the reservation is cancelled while the reservation is waiting 1 -- normal, 0 -- fault
M341	Reservation 1 of station fault	Input	
M342	Reservation 2 of station fault	Input	
M343	Reservation 3 of station fault	Input	
M344	Reservation 4 of station fault	Input	
M345	Reservation 5 of station fault	Input	
M346	Reservation 6 of station fault	Input	
M347	Reservation 7 of station fault	Input	
M348	Reservation 8 of station fault	Input	
M349	Reservation 9 of station fault	Input	

M350	Run button of reservation 1, input	Input	M35x valid condition: rising edge, keep the set valid time signal valid (operation parameter - 20 #), falling edge
M351	Run button of reservation 2, input	Input	
M352	Run button of reservation 3, input	Input	
M353	Run button of reservation 4, input	Input	
M354	Run button of reservation 5, input	Input	When there is no reservation at the station, increase the reservation.
M355	Run button of reservation 6, input	Input	
M356	Run button of reservation 7, input	Input	
M357	Run button of reservation 8, input	Input	When the station is suspended, continue to run.
M358	Run button of reservation 9, input	Input	
M359	Run button of reservation 10, input	Input	When the station is waiting for an appointment, cancel the appointment.
M360	Running light of reservation 1, output	Output	Operation parameter - 30 #, set m36x output mode: 0 - the light is on during operation, otherwise it stops off.
M361	Running light of reservation 2, output	Output	
M362	Running light of reservation 3, output	Output	1-program completion output (2 seconds) 2 - the light is on when making an appointment and off when completing
M363	Running light of reservation 4, output	Output	
M364	Running light of reservation 5, output	Output	
M365	Running light of reservation 6, output	Output	No appointment: 0-off 1-Off 2-off
M366	Running light of reservation 7, output	Output	Reservation waiting: 0-flashing 1-Off 2-off
M367	Running light of reservation 8, output	Output	Reservation operation: 0 - on 1 - Off 2 - on
M368	Running light of reservation 9, output	Output	
M369	Running light of reservation 10, output	Output	During reservation stop: 0-off, 1-on after operation completion, 2-on after two seconds

M370	Stop button of reservation 1, input	Input	When the corresponding station is running, it stops running. No action in other states
M371	Stop button of reservation 2, input	Input	
M372	Stop button of reservation 3, input	Input	
M373	Stop button of reservation 4, input	Input	
M374	Stop button of reservation 5, input	Input	
M375	Stop button of reservation 6, input	Input	
M376	Stop button of reservation 7, input	Input	
M377	Stop button of reservation 8, input	Input	
M378	Stop button of reservation 9, input	Input	
M379	Stop button of reservation 10, input	Input	
M380	Stop light of reservation 1, output	Output	The stop light is on when the appointment is stopped
M381	Stop light of reservation 2, output	Output	
M382	Stop light of reservation 3, output	Output	
M383	Stop light of reservation 4, output	Output	
M384	Stop light of reservation 5, output	Output	
M385	Stop light of reservation 6, output	Output	
M386	Stop light of reservation 7, output	Output	
M387	Stop light of reservation 8, output	Output	
M388	Stop light of reservation 9, output	Output	
M389	Stop light of reservation 10, output	Output	
M392	J9 axis out of tolerance detection alarm switch	Input	1: J9-J12 axis out of tolerance detection is prohibited; 0: out of tolerance detection is allowed;
M393	J10 axis out of tolerance detection alarm switch	Input	
M394	J11 axis out of tolerance detection alarm switch	Input	
M395	J12 axis out of tolerance detection alarm switch	Input	
M396	Return to zero state relay of J9 axis	status	1: Return to zero state relay of J9-J12 axis
M397	Return to zero state relay of J10 axis	status	
M398	Return to zero state relay of J11 axis	status	
M399	Return to zero state relay of J12 axis	status	
M470	drive alarm of J13 axis	Input	1: drive alarm of J13-J16 axis
M471	drive alarm of J14 axis	Input	
M472	drive alarm of J15 axis	Input	
M473	drive alarm of J16 axis	Input	

M474	Drive of J13 axis is ready	Input	1: Drive of J13-J16 axis is ready
M475	Drive of J14 axis is ready	Input	
M476	Drive of J15 axis is ready	Input	
M477	Drive of J16 axis is ready	Input	
M478	Enable completion of J13 axis	Input	1: Enable completion of J13-J16 axis
M479	Enable completion of J14 axis	Input	
M480	Enable completion of J15 axis	Input	
M481	Enable completion of J16 axis	Input	
M482	J13 axis interference	status	1: J13-J16 axis enters the interference area; 0: outside the interference area;
M483	J14 axis interference	status	
M484	J15 axis interference	status	
M485	J16 axis interference	status	
M486	J13 axis out of tolerance detection alarm switch	Input	1: Out of tolerance detection of j13-j16 axis is prohibited; 0: out of tolerance detection is allowed;
M487	J14 axis out of tolerance detection alarm switch	Input	
M488	J15axis out of tolerance detection alarm switch	Input	
M489	J16 axis out of tolerance detection alarm switch	Input	
M490	Return to zero state relay of J13 axis	status	1: Return to zero state relay of J13-J16 axis
M491	Return to zero state relay of J14 axis	status	
M492	Return to zero state relay of J15 axis	status	
M493	Return to zero state relay of J16 axis	status	
M494	J11 axis interference	status	1: J11-j12 axis enters the interference area; 0: outside the interference area;
M495	J12 axis interference	status	

For user's PLC adjustment, need to add M332-M334 (station allowed, required), M342-M344 (station failure, required), M352-M354 (station start, required), M372-M374 (station stop, required), M362-M364 (running light, as required), M382-M384 (stop light, as required) need to be added.

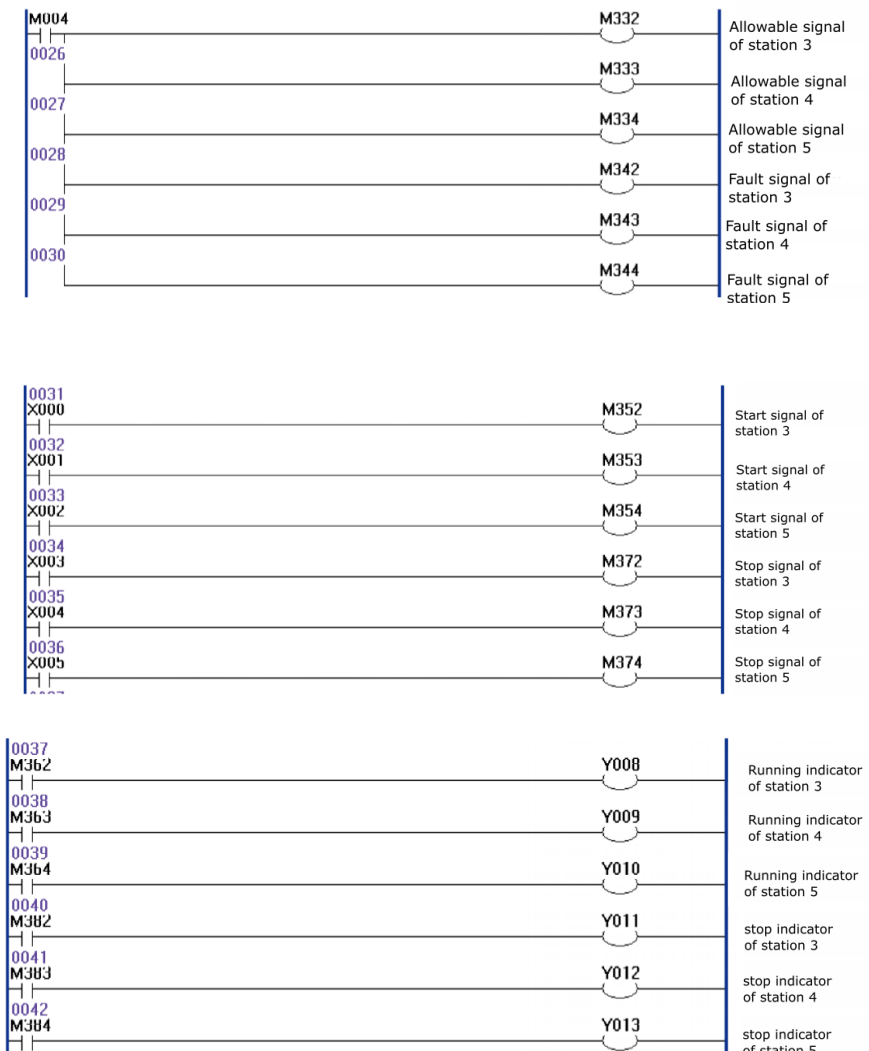


Fig.5.3.4

In the wiring diagram below, X00-X05 and Y08-Y13 are connected to the I / O board (used by the user), and TX1 and TX2 terminals are defined as 10, 11, 12 and 13 in series. ST4-1, ST4-2, ST4-3 and ST4-4 are connected to the safe board interface and terminal J13.

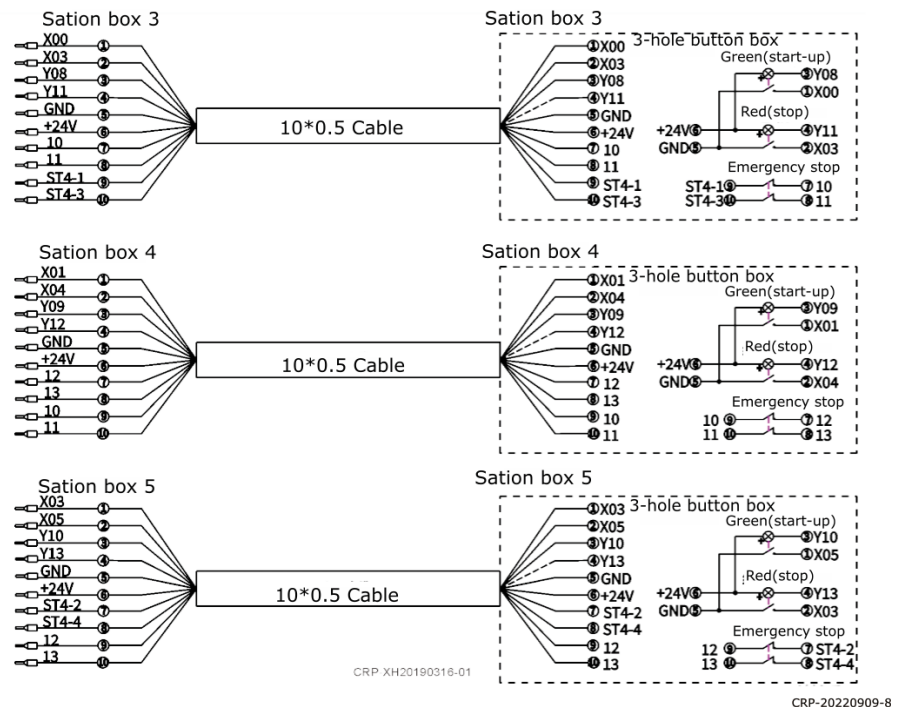


Fig.5.3.5

5.4 Description of expansion axis (CDK □)

5.4.1 Introduction

Present, the CRP-G4-CD60 electric cabinet can be expanded by two axes. The maximum rated current of the expanded axis module is 15A. When the motor exceeds 15A, it cannot reach the rated load.

When adding the expansion axis, you also need to adjust:

1. Adjust the brake power supply (UK2) from 100W to 150W, otherwise the brake cannot be opened effectively. For the leaving factory electric cabinet with expansion shaft, the band brake power supply has been adjusted to 150W.
2. Transformer (TC1) is adjusted from 5000VA to 8000VA. For the leaving factory electric cabinet with expansion shaft, the transformer has been adjusted to 8000VA.

5.4.2 Installation and connection of expansion axis

1. Turn off the main power supply (SQ1) of the control cabinet and turn off the front-end power switch input to the control cabinet.
2. Replace the band brake power supply (UK2).
3. Replace the transformer (TC1).
4. Install the expansion axis (CDK1 or CDK2) to the reserved position of the electrical cabinet, and connect the interconnection cable (X03-X04, X05-X06, X07-X08) with CD60-CRX8. Please refer to the following figure (see the attachment for the wiring diagram of expansion 9 and 10 shafts).

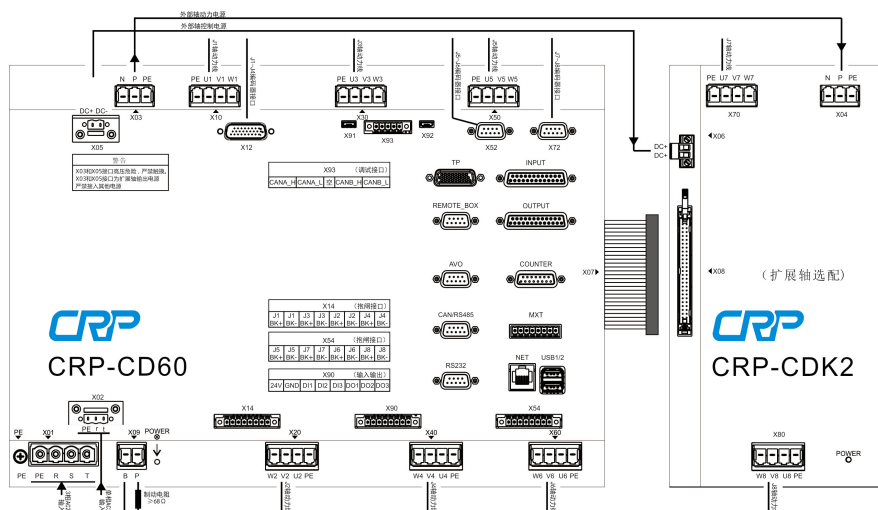


Fig.5.4.1

5. Connect the expansion axis motor. The motor encoder line is connected to the X72 terminal of CD60-CRX8. The power line of the expansion axis motor is connected to the X70 (7-axis) and X80 (8-axis, if any) terminals of CDK □.

Refer to [CRP-CD60-CRX8 System Manual](#) and [CRP-CD60-CRX8 Commissioning Manual](#) for port details.

Refer to [CRX8 External Axis Extension Manual](#) for the commissioning instructions of the expansion axis.

6. Maintenance of Electric Cabinet

6.1 Daily maintenance



WARNING

Please read carefully and operate after you have understood this section!

Misoperation may be dangerous, and moderate or minor injury accidents and equipment failures may occur.

Working principle:

1. Pay attention to follow the instructions of the detergent manufacturer when cleaning.
2. The cleaning agent must be prevented from penetrating into the electrical components.
3. Compressed air is not allowed for cleaning.
4. Do not spray with water.

Cleaning steps:

1. Loosen and suck out the accumulated dust.
2. Clean the robot control system with a rag soaked with flexible detergent.
3. Clean cables, plastic parts and hoses with detergent without dissolving agent.
4. Replace the damaged or illegible written instructions and nameplates, and supplement the missing instructions and nameplates. The following daily maintenance must be carried out.

M a i n t e n a n c e position	Inspection items	repair time	remarks
G4 control cabinet	Check whether the electric cabinet door is completely closed	Every day	
	Check whether the structural gap inside the seal is damaged	Every month	
Cooling fan on the back	Confirm action	suitable	When the power is turned on
Emergency stop button	Confirm action	suitable	When the servo is turned on
Battery	Confirm action	suitable	In teaching mode
Power supply	Confirm action	suitable	
Circuit breaker wire	Confirm whether the wire falls off, loosens or breaks, and confirm the voltage	suitable	

★ Note
Regular maintenance can ensure: • continuous operation of robots to improve production efficiency. • improve the service life of the robot. • improve MTBF • avoid the trouble caused by maintenance

6.2 Mainboard battery replacement

The battery of the controller motherboard needs to be replaced after a certain period of use.

1. Open the battery cover on the drive control integration.
2. Loosen the battery plug for replacement.

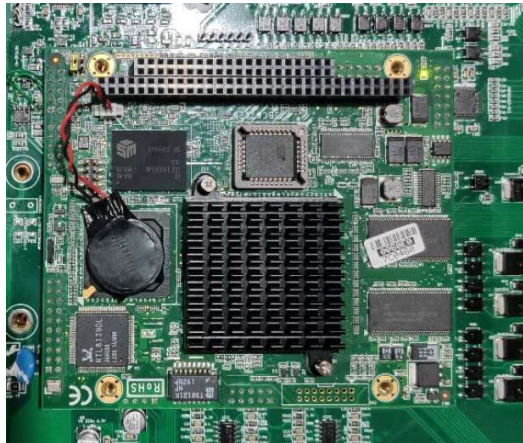
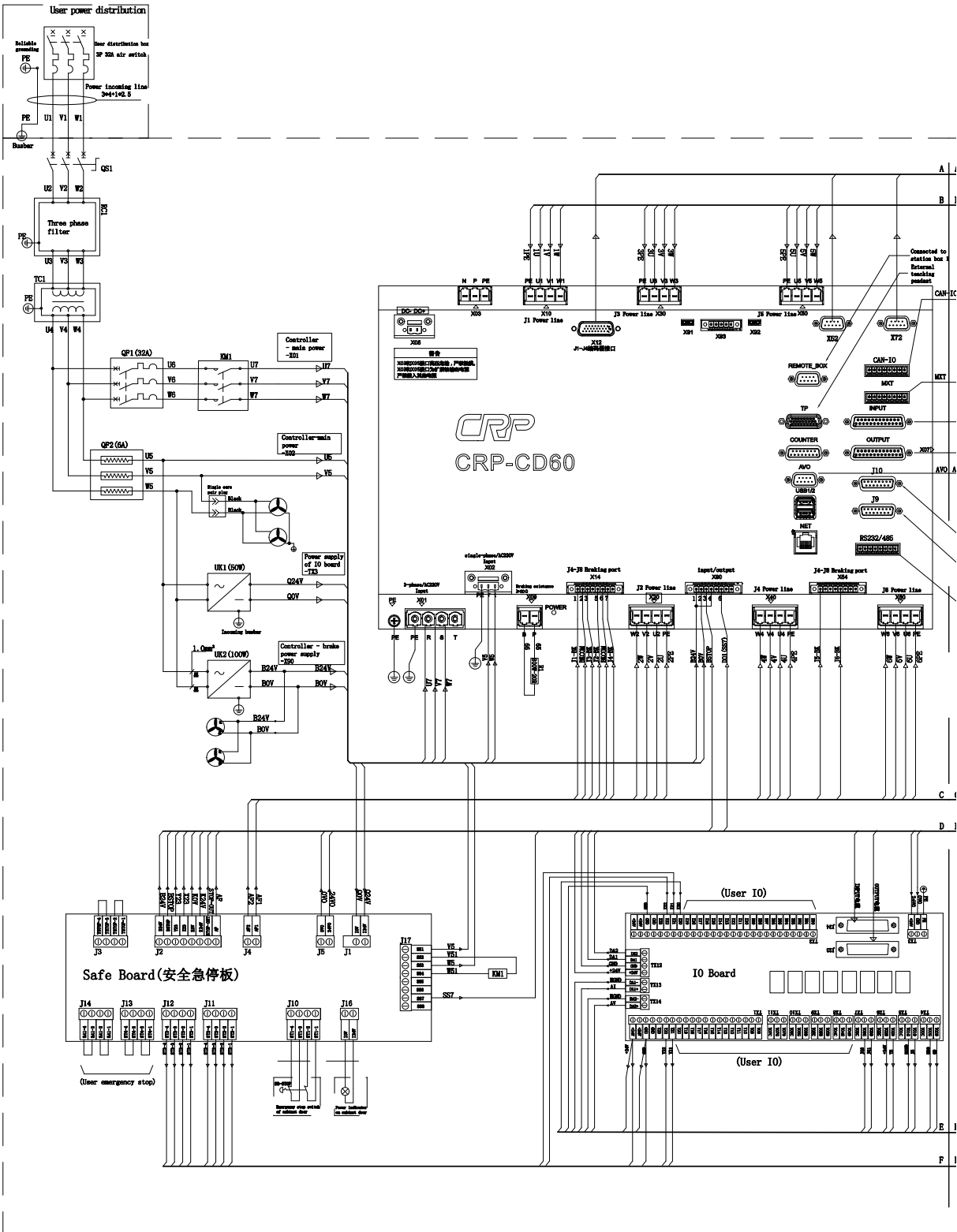


Fig.6.2.1

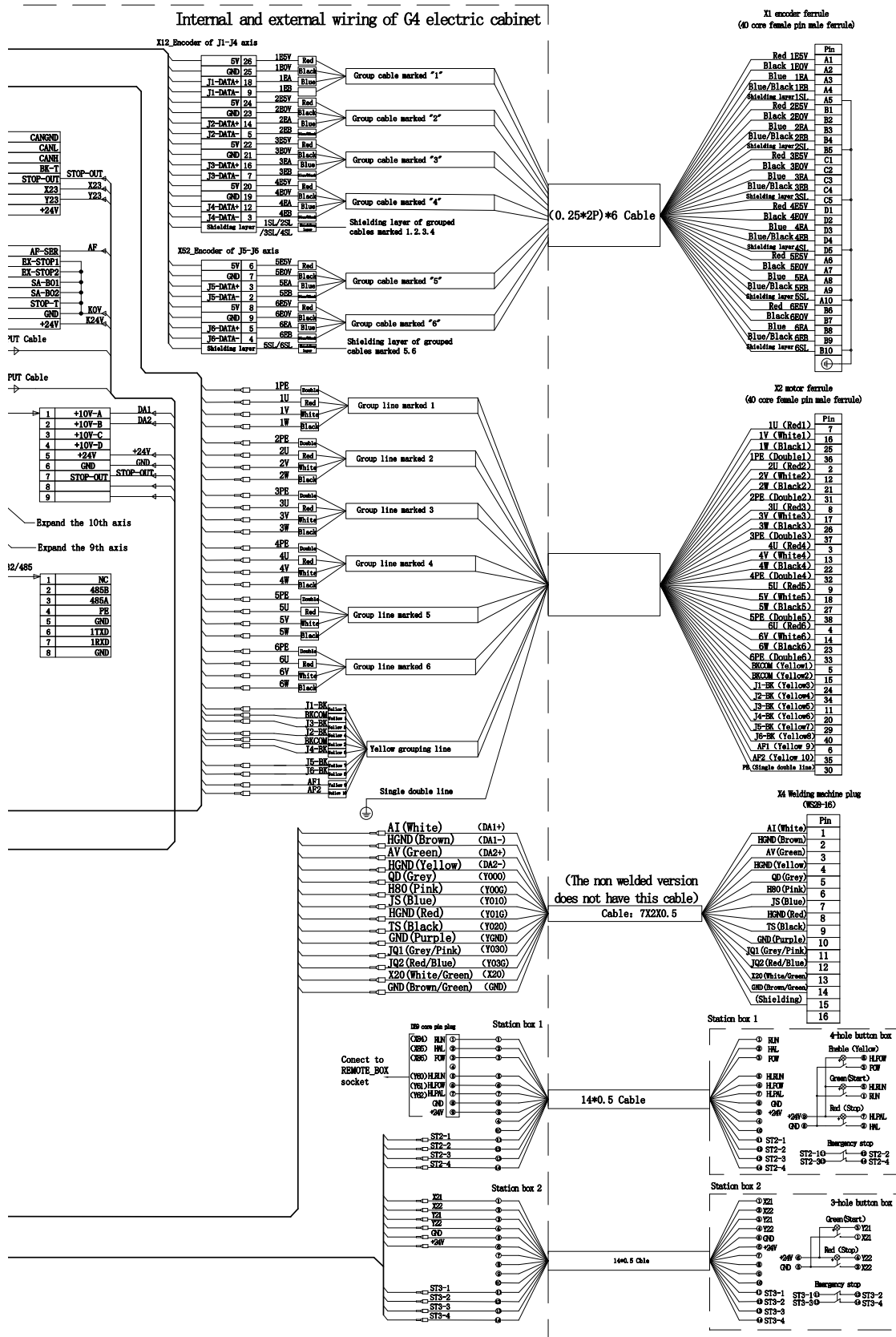
3. Insert the new button battery.
4. Re lock the battery cover screw.

Appendix

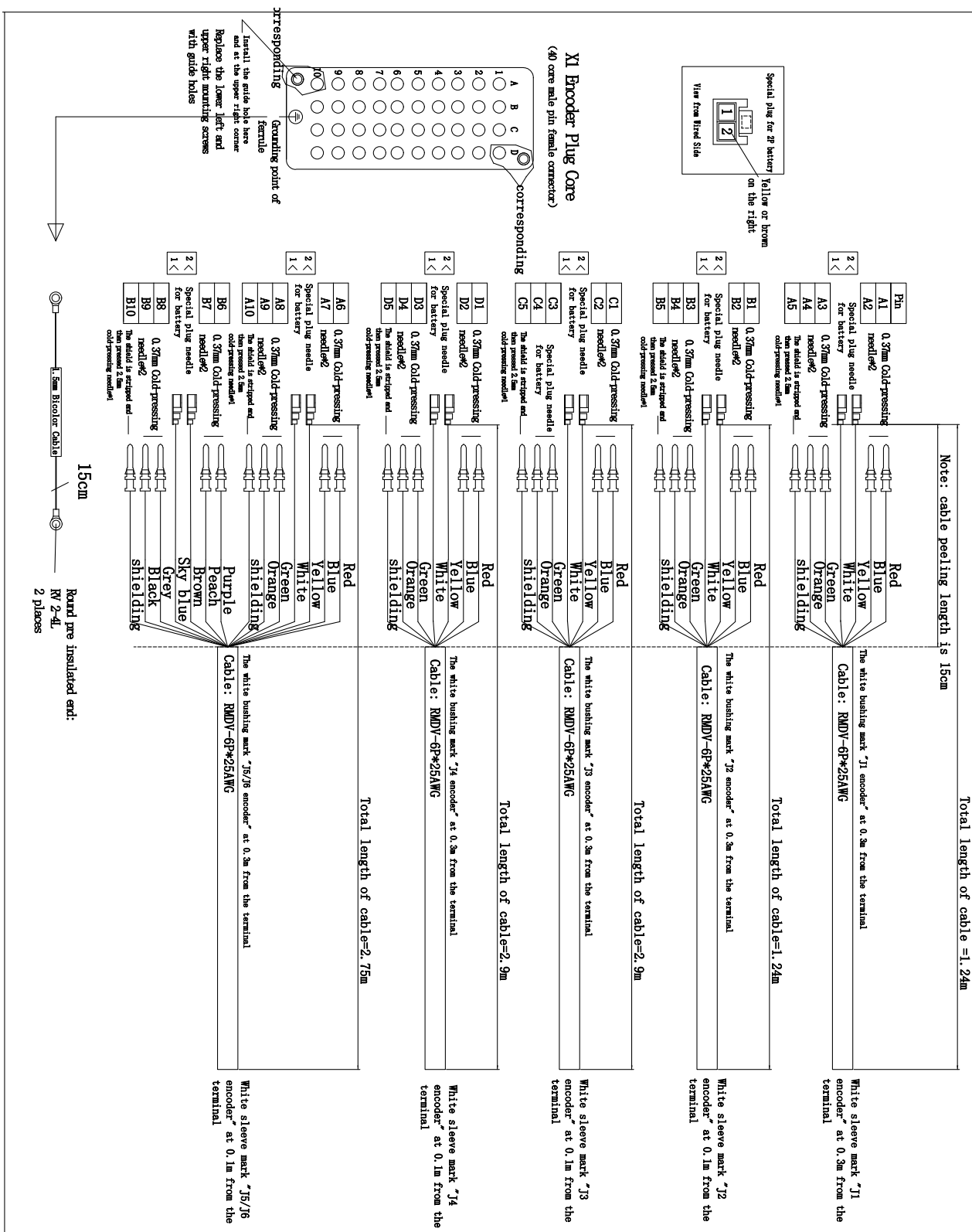
Attachment 1: electrical schematic diagram



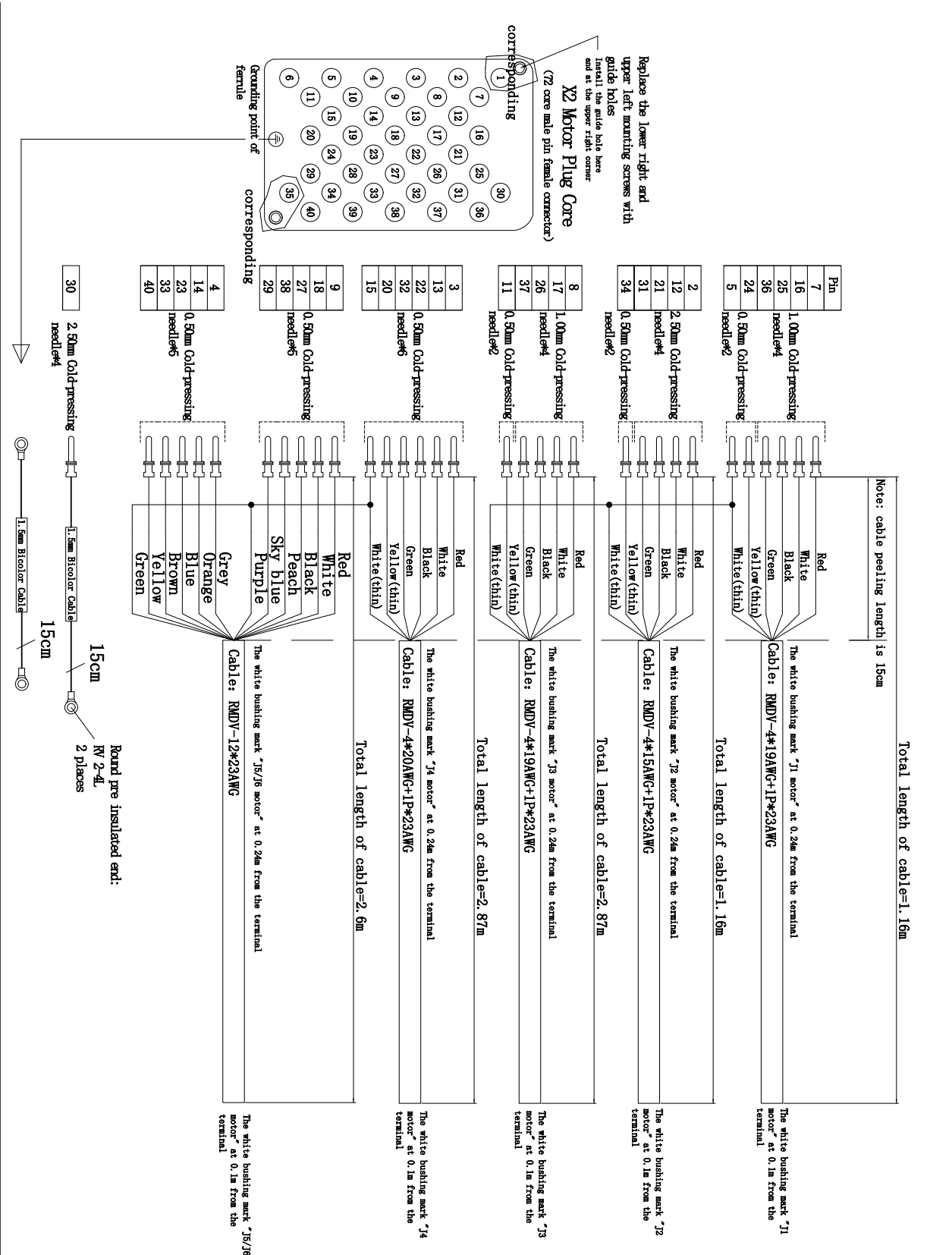
Internal and external wiring of G4 electric cabinet



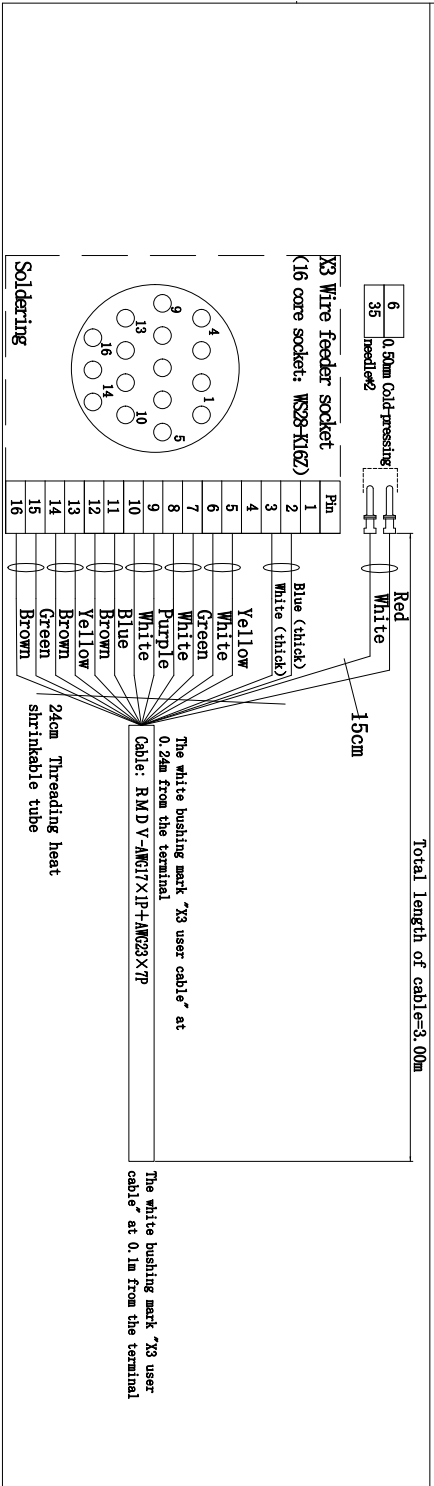
Attachment 2: X1 encoder cable connection diagram



Attachment 3: X2 motor cable connection diagram



Attachment 4: X3 welding connection diagram (welding version)





WeChat public account



Tiktok



Reference material

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